Resampling Cannot See the User: Detecting User-Specific Hallucination in Interactive Agents with Probabilistic Commitments

Anonymous submission

Abstract

An agent serving an individual must act on facts about that person it was never told: whether a scan is safe today, whether a sedative is needed. When it asserts such facts confidently and wrongly, nothing external can check it, because the truth lives with the user. We show that sampling-consistency detectors cannot tell users apart: conditional on the conversation, the model's resamples carry no information about which user it is wrong about, and the more consistently an agent fabricates, the closer their AUROC is to $1/2$. We propose Probabilistic Game Environments (PGE), a protocol in which the agent states each assumption as an explicit commitment with a confidence and the user's implicit reactions are recorded, and a label-event posterior that turns commitments and reactions into per-commitment and per-session risk. In a simulator where the truth is known, exact Bayes predictors split what is detectable into population risk, the chance that these commitments are wrong for a user with the same intake, and individual evidence from the user's reactions. In a pre-registered evaluation on fresh sessions with five LLM agents from four families, sampling detectors are near chance on user-specific hallucination (SelfCheckGPT 0.566 [0.535, 0.595] AUROC pooled), while detectors that read commitments and reactions are well above it. The user's reactions add +.072 [+.054, +.090]* AUROC to our posterior and, at the Bayes level, +.087 [+.069, +.107]* beyond the population risk of the same commitments. Our posterior is the most label-efficient detector, but with hundreds of labelled sessions a supervised model on the same inputs overtakes it. [PENDING: clarification headline] We report every pre-registered outcome, including those against our method.

Keywords: LLM agents; hallucination detection; uncertainty; clarification; implicit feedback.

1 Introduction

An agent that schedules imaging appointments must decide whether a patient may have a contrast scan today. The intake form says the patient had a recent treatment, but not when. A careful agent would say it does not know; many instead answer "safe, $p = 0.85$" and proceed. We call this a *user-specific hallucination*: a confident assertion about the current user whose truth depends on something the user never disclosed. It differs from the factual hallucinations that dominate the literature [10]: there is no document or search engine to check it against.

The leading black-box detectors resample the model and flag answers it does not reproduce [8, 16, 20]. That presumes hallucinations are stochastic. Fabricated user facts need not be: an underdetermined question maps the same context to the same guess every time. More fundamentally, two users who share a conversation (the same intake answers and, where the agent is shown them, the same reactions) are indistinguishable to the model, so nothing computed from the model and its context can tell which of them it is wrong about (Theorem 1). Context can still reveal *population* risk, for instance that "no sedation needed" is wrong for a third of patients, and we show that a well-calibrated context-only predictor can exploit it. Sampling-consistency scores of a self-consistent agent do not even do that (Corollary 1).

The way out is to look at the user rather than the model. In PGE the agent must externalize each assumption as a *commitment* $(\text{attribute}, \text{value}, p)$, and the user's reaction (accepting it, contradicting it, or ignoring it) is recorded. Reactions are noisy and partial, but they depend on the truth for *this* user, which no resample does: under a simple model the posterior of the hallucination event uses them exactly (Proposition 1), and less noisy reactions cannot hurt (Theorem 2). Our contributions:

1. **Why sampling fails, measured.** A within-context impossibility result and a dispersion bound, checked per commitment on five agents: on commitments whose truth depends on undisclosed user facts, resample disagreement predicts error at 0.517–0.555 AUROC across agents, a single reaction with 25% noise at 0.744–0.746.

2. **A decomposition of detectability.** Exact Bayes predictors in a simulator with known truth separate the population risk of the commitments from the individual evidence in the reactions; the posterior given commitments and reactions bounds what any detector, with or without resamples, can achieve.

3. **A protocol and a posterior.** PGE's commitments and reactions make user-specific hallucination detectable; its label-event posterior gives per-commitment risk

that is exact under a conditional-independence model (estimated by plug-in), is the most label-efficient detector we test, and tells the agent what to ask the user, which we measure with the agent re-reasoning after each answer.

4. **A pre-registered evaluation.** After an independent audit of our development results, we fixed the analysis, fixed a protocol bug, and generated fresh sessions with five agents from four families and a natural-language simulated patient. We report every registered hypothesis, including those that go against the posterior.

2 Related Work

Hallucination detection. Sampling-based detectors measure agreement across resamples [8, 16, 20]; semantic entropy is explicitly scoped to stochastic confabulations, not consistent errors [8]. Learned confidence models from a model's generations [31], white-box probes [2, 4, 15], and verification by the model's own questions [7] or by search [32] are all functions of the model given its context, so Theorem 1 applies to them. Consistent errors are documented in snowballing [35] and self-contradiction studies [21], and calibrated models must hallucinate on facts absent from training [12]; our setting is the per-user analogue.

Hallucination in agents. MIRAGE-Bench [36] taxonomizes agent actions unfaithful to instructions, history or observations; CAR-bench [14] tests whether agents admit missing information with a simulated user; AgentHallu [18] localizes the step that introduced an error. We study detection when the missing information is the user's own.

Clarification. When to ask has been framed through expected value of perfect information [25, 30], ambiguity detection [34], conformal prediction [26], and information-gain rewards on τ-bench [6, 33]. These reason about the agent's uncertainty before it acts; we use post-hoc commitment risk to choose which assertions to confirm, and measure the effect with the agent re-reasoning.

Implicit and noisy feedback. Implicit signals such as clicks are informative but biased [11]. Our estimation step inverts a noise channel, as in learning from noisy labels [1, 22, 28] and annotator models [5]. Agents that revise under user pushback raise sycophancy concerns [29]; our simulated user follows a long line of user simulation for dialogue [27].

3 Problem, Protocol, and Labels

Definition 1 (User-specific hallucination)**.** *A* commitment *asserts a value v for an attribute a of the current user with confidence* $p \in [0,1]$. *It is* user-dependent *if its true value is not determined by what the user has disclosed, and it is a user-specific hallucination if it is user-dependent, confident ($p \geq \tau$), and wrong.*

Protocol. A PGE session has four stages. (i) *Intake*: the user answers a few coarse questions; the agent sees only these answers. (ii) *Commitment*: at each turn the agent emits one commitment in a fixed format, `[attribute]: value, p=confidence`. (iii) *Reaction*: the user's implicit reaction $F_t \in \{-1, 0, +1\}$ (contradicts, no reaction, accepts) is recorded and shown to the agent for its next six turns. (iv) *Detection and intervention*: a detector scores the session and its commitments; high-risk commitments trigger a confirmation question.

Labels. With confident set $H = \{t : p_t \geq \tau\}$, $\tau = 0.6$, the *composite* label L (Eq. (1)) marks a session in which at least half of the confident commitments are wrong, and is 0 if there are none. It mixes two kinds of error. Some attributes are determined by the intake through rules the agent was never shown (e.g. contrast is needed iff urgency is medium or high); errors there are guesses about the clinic, not about the user. The *user-specific* label L_U applies the same rule to the confident commitments that are user-dependent in that session, and is 0 when there are none. Whether a commitment is user-dependent is decided per session by enumerating the hidden profile's support: `scan_safe_now`, for example, depends on days since treatment only when a treatment was recent. L_U is our primary label; L is reported as secondary.

4 What Resampling and Feedback Can Reveal

Setup. A user u is drawn from a population. At turn t the agent conditions on a context X_t: the coarse intake $\phi(u)$, the question, and the interaction history, which in PGE displays earlier reactions; X denotes $(X_1, \ldots, X_T)$. It emits a commitment $c_t = (a_t, v_t, p_t)$ (attribute, value, reported confidence) from its policy $\pi(\cdot \mid X_t)$. Apart from the truth and the reactions, nothing depends on u except through X. The truth $y_a(u)$ is private to the user; the commitment is wrong if $w_t = \mathbf{1}[v_t \neq y_{a_t}(u)]$. With confident set $H = \{t : p_t \geq \tau\}$ and error fraction ρ, a session hallucinates iff

$$L = \mathbf{1}\Big[\, |H| \geq 1 \text{ and } \textstyle\sum_{t \in H} w_t \;\geq\; \lceil \rho\, |H| \rceil \Big]. \qquad (1)$$

The results below hold for any binary label determined by (C, u): a single w_t, or L_A, Eq. (1) with H replaced by $H_A = \{t \in H : a_t \in A\}$ for an attribute set A determined by X (for the user-specific label L_U, the attributes the intake does not determine). We write L.

A *context-and-sample detector* is any $S = h(X, C, Z, \xi)$ with ξ independent randomness and Z drawn, given (u, X, C), from a kernel $K(\cdot \mid X, C)$ that does not involve

u: resamples at any temperature, token log-probabilities, hidden states. A *feedback detector* also sees reactions $F_t \in \{-1,0,+1\}$ (contradicts, none, accepts). By the Neyman–Pearson (NP) lemma [23], among all, possibly randomized, detectors that are functions of an observation O, thresholding $P(L=1 \mid O)$ (randomized at ties) has the highest true-positive rate at every false-positive rate, hence the highest AUROC; ties count $\frac{1}{2}$. Theorem 1 is the standard data-processing consequence of NP [3, 23], stated for this setting.

Theorem 1 (Self-consistency bound)**.** *Let $\eta_\varnothing = P(L = 1 \mid X,C)$ and S be a context-and-sample detector. Given (X,C), S is independent of u and L, so within a context it ranks hallucinating above clean sessions with probability exactly $\frac{1}{2}$. Across contexts, its ROC curve lies pointwise below that of the Bayes predictor $\eta_\varnothing$.*

Proof. Given (u,X,C), S depends only on (Z,ξ), whose law does not involve u, and L is a function of (C,u). So the law of S given (X,C) does not depend on L, $P(L=1 \mid X,C,S) = \eta_\varnothing$, and NP applies to $O=(X,C,S)$. □

Since X contains the reactions the agent was shown, $\eta_\varnothing$ is not feedback-free, and detectors built from (X,C), such as context-only classifiers, can reflect them and exploit base rates across contexts. Resampling cannot add to (X,C) or separate users who share a context; a feedback detector reads the reactions directly, including the last. For dispersion scores, more is true:

Corollary 1 (Dispersion under systematic fabrication)**.** *Let S be a score of the commitments and their k resamples (agreement rate, semantic entropy, SelfCheck-style inconsistency) that takes a fixed value $s^\star$ whenever every resample equals its commitment, and $e_\ell = P(S \neq s^\star \mid L = \ell)$. Then*

$$\left|\mathrm{AUROC}(S) - \tfrac{1}{2}\right| \;\le\; \tfrac{1}{2}\big(1-(1-e_0)(1-e_1)\big) \;\le\; \tfrac{1}{2}(e_0+e_1).$$

If $s^\star$ is the least suspicious value of S, then $\left|\mathrm{AUROC}(S) - \frac{1}{2} - \frac{1}{2}(e_1-e_0)\right| \le \frac{1}{2}e_0e_1$. If every turn's resampling distribution is a point mass at the committed value, $\mathrm{AUROC}(S) = \frac{1}{2}$ however often the commitments are wrong.

Proof. For independent positive and negative draws, both scores equal $s^\star$ with probability $(1-e_0)(1-e_1)$; such pairs are ties and contribute exactly $\frac{1}{2}$. All other pairs contribute a value in $[0,1]$. If $s^\star$ is least suspicious, a pair where only the positive (negative) differs from $s^\star$ contributes 1 (0); only pairs where both differ, with probability e_0e_1, are free. □

On the development corpora (§7.1), the 70B agent's user-dependent commitments (label w_t) have $e_1 = 0.079$ and $e_0 = 0.078$: the corollary caps disagreement's AUROC at 0.576, and its sharp form pins it within $\frac{1}{2}e_0e_1$ of $\frac{1}{2} + \frac{1}{2}(e_1 - e_0) \approx \frac{1}{2}$; we measure 0.500, against 0.764 for one noisy reaction. For the less self-consistent 8B agent the cap (0.884) is loose.

Theorem 2 is Blackwell's garbling theorem [3] for a symmetric channel. Let F^ε be feedback generated for the detector: given (u,X,C), which F_t^ε are 0 has a law depending only on (X,C), and each other F_t^ε is $1-2w_t$ with its sign flipped independently with probability ε. Let $\eta_\varepsilon = P(L=1 \mid X,C,F^\varepsilon)$ ($\varepsilon = 0$ is perfect feedback).

Theorem 2 (Less noisy feedback cannot hurt)**.** *Fix the joint law of (u,X,C); in particular, it does not depend on ε. For $0 \le \varepsilon \le \varepsilon' \le \frac{1}{2}$ the ROC curve of η_ε lies pointwise above that of $\eta_{\varepsilon'}$, which lies above that of $\eta_\varnothing$, and $\eta_{1/2} = \eta_\varnothing$ almost surely.*

Proof. If $\varepsilon' = \varepsilon$ there is nothing to show. Otherwise $\varepsilon < \frac{1}{2}$, and flipping each nonzero F_t^ε independently with probability $\delta = (\varepsilon' - \varepsilon)/(1-2\varepsilon) \in [0, \frac{1}{2}]$ yields flip rate $\varepsilon + \delta(1-2\varepsilon) = \varepsilon'$, hence the joint law of $(u,X,C,F^{\varepsilon'})$. So every detector using $F^{\varepsilon'}$ is a randomized detector using F^ε, and NP gives the first inequality; ignoring F gives the second. At $\varepsilon = \frac{1}{2}$, F^ε is independent of u given (X,C): whether a reaction occurs depends only on (X,C), and every observed sign is a fair coin. □

The premise is essential: if the agent conditions on the same reactions, the law of (X,C) changes with ε and monotonicity can fail. In a two-turn example where the agent copies the reaction to an unconfident answer into a confident one on a correlated attribute without feedback, $\mathrm{AUROC}(\eta_\varepsilon)$ is 0.50 at $\varepsilon = 0$ and 0.71 at $\varepsilon = 0.25$. In PGE, Theorem 2 thus covers regenerating feedback for the detector with the agent's trajectories fixed.

The label-event posterior. Let $\kappa_t = (a_t, v_t, b(p_t))$ be the detector-visible *cell*, where $b(\cdot)$ buckets confidence, and $\mathscr{H}_t = (X_t, c_{1:t}, F_{<t}, w_{<t})$ the history up to the t-th commitment with the earlier errors. (A1) Given $\mathscr{H}_t$, $w_t \sim \mathrm{Bernoulli}(\pi_{\kappa_t})$. (A2) Given $(\mathscr{H}_t, w_t)$, $F_t = 0$ with a probability that depends only on a_t, and otherwise $F_t = 1 - 2w_t$ with its sign flipped with probability ε. Up to a factor free of w_t, the likelihood of F_t is $\ell_{w_t}(F_t)$ with $\ell_w(0) = 1$, $\ell_1(-1) = \ell_0(+1) = 1-\varepsilon$ and $\ell_1(+1) = \ell_0(-1) = \varepsilon$. (A1) excludes error cascades and intake information beyond the cell, but lets the agent choose cells after seeing feedback.

Proposition 1 (Exact posterior)**.** *Under (A1)–(A2), $P(w_{1:T} \mid X,C,F) = \prod_t q_t^{w_t}(1-q_t)^{1-w_t}$ with $q_t = \pi_{\kappa_t}\ell_1(F_t)/\big(\pi_{\kappa_t}\ell_1(F_t) + (1-\pi_{\kappa_t})\ell_0(F_t)\big)$. Hence $\eta_\varepsilon = P(L=1 \mid X,C,F)$ is 0 if $H = \varnothing$ and otherwise the Poisson-binomial tail $P\big(\sum_{t\in H} B_t \ge \lceil \rho|H| \rceil\big)$, $B_t \sim \mathrm{Bernoulli}(q_t)$ independent, computable in $O(|H|^2)$. It is Bayes-optimal under (A1)–(A2) with the true priors and channel; the implementation plugs in estimates.*

Proof. Write the joint law in the order $c_1, w_1, F_1, c_2, \ldots$. The policy and the questions see u only through X, so $p(X,C,F,w) = A(X,C,F)\prod_t \pi_{\kappa_t}^{w_t}(1-\pi_{\kappa_t})^{1-w_t}\ell_{w_t}(F_t)$, where A collects the intake prior, the questions, the policy and the w-free factors of (A2). Normalizing over $w \in \{0,1\}^T$ gives independent $\mathrm{Bernoulli}(q_t)$. H is fixed

by C, L depends on w only through $\sum_{t\in H} w_t$, and NP gives optimality. □

The same proof gives the posterior of L_A (the tail over H_A), and holds for any channel $P(F_t \mid \mathscr{H}_t, w_t) = \ell_{w_t}(F_t)$ that may depend on a_t, as in the general-channel detector. Under (A1) this user's reactions enter only through $\ell_1(F_t)/\ell_0(F_t)$; setting these to 1 leaves the population base rates π_κ (the prior-only ablation). Cascades violate (A1), but a simple cascade model does not help (§7.7).

Proposition 2 (Priors are identified where feedback exists)**.** *Assume (A2) and let $\pi_\kappa = P(w_t = 1 \mid \kappa_t = \kappa)$. Let n_κ count the commitments in cell κ that received a reaction ($F_t \neq 0$) and $\hat{r}_\kappa$ the fraction contradicted, and assume their errors are independent given n_κ (implied by (A1), or by i.i.d. sessions in which each attribute is committed at most once). Then the contradiction rate is $r_\kappa = \varepsilon + (1-2\varepsilon)\pi_\kappa$, and for known $\varepsilon < \frac{1}{2}$, $\hat{\pi}_\kappa = (\hat{r}_\kappa - \varepsilon)/(1-2\varepsilon)$ needs no labels and, given n_κ, with probability at least $1-\delta$ [9],*

$$|\hat{\pi}_\kappa - \pi_\kappa| \;\le\; \frac{1}{1-2\varepsilon}\sqrt{\frac{\log(2/\delta)}{2n_\kappa}};$$

projecting onto $[0,1]$ does not increase the error. Cells on attributes that never receive feedback are not identified. If ε is unknown, the reactions depend on (π,ε) only through $(r_\kappa)_\kappa$, so every $\varepsilon' \in [0, \frac{1}{2})$ with $\varepsilon' \le \min_\kappa \min(r_\kappa, 1-r_\kappa)$, with $\pi'_\kappa = (r_\kappa - \varepsilon')/(1-2\varepsilon')$, fits equally well.

One scalar anchor (ε, or one cell's $\pi_\kappa \neq \frac{1}{2}$) resolves this; we choose ε by training-split label likelihood. Unidentified cells get the bucket prior pooled over the other attributes. The bound covers the raw estimate, not the implementation's clipping to $[0.01, 0.99]$ and shrinkage toward coarser cells (value → attribute → bucket).

5 Detectors

PGE posterior. The score is a plug-in estimate of the posterior of Proposition 1. Each confident commitment is assigned to a cell $\kappa = (\text{attribute}, \text{value}, \text{confidence bucket})$ with buckets $[0.6, 0.65)$, $[0.65, 0.75)$, $[0.75, 1]$. Cell priors are estimated from training reactions by the method of moments (Proposition 2), shrunk toward attribute-level and bucket-level estimates (strength $k = 5$); cells on the two attributes that never receive reactions are not identified and get the pooled bucket-level prior. Our pre-registered primary detector, V6, uses a general channel: the contradiction rates $r_w = P(F = -1 \mid w)$ are chosen on training labels over a grid ($r_1 > r_0 + 0.1$), priors are $\pi = (\text{rate} - r_0)/(r_1 - r_0)$, and the confirmation rates $g_w = P(F = +1 \mid w)$ are fit by least squares; the symmetric flip channel (V3, with flip rate ε chosen on training labels) is the special case $r_0 = g_1 = \varepsilon$, $r_1 = g_0 = 1 - \varepsilon$. Proposition 1 covers this channel, since its proof uses only $P(F_t \mid w_t)$; the channel scale is chosen on training labels because reactions alone do not identify it (Proposition 2). For L_U the label event is restricted to commitments on attributes that depend on a hidden field anywhere in the schema, a set the detector can know without the clinic's rules. V6 is therefore the posterior of a proxy event (L_A of Section 4 with A the schema-level set), not of L_U: on the development corpora the two events differ in most sessions and the labels disagree on a fifth to a quarter of them. *Prior-only* (V6−prior) keeps the fitted priors and ignores the test user's reactions; V6 minus V6−prior is the feedback increment.

Baselines. SelfCheckGPT and semantic entropy on $k = 3$ resamples per turn at temperature 0.9 (exact-match clustering, adequate for categorical answers; for L_U computed on the user-dependent turns); Chain-of-Verification with the agent as its own verifier [7]; mean stated confidence (sign chosen on the training split); two feedback statistics (fraction of contradicted confident commitments, fraction of contradicted turns); a logistic regression on 14 reasoning-graph features; a two-layer GCN [13]; five block-model graphon likelihoods [19], of which we report the best on test, favouring the baseline. Two supervised baselines see exactly the posterior's inputs: *ctx-LR* is an L_2 logistic regression ($C = 0.1$, fixed) on counts of confident commitments per (attribute, value) cell, $|H|$, the number of turns and mean confidence, with no reactions and no samples; *ctx-LR+F* adds counts per (attribute, value, reaction) cell.

6 Experimental Design

Environment. A simulated imaging clinic. Five coarse intake answers are shown to the agent; four hidden fields (days since treatment, allergy severity, kidney function, anxiety) are not. Each session asks for 6–12 of 12 attributes in dependency order. Six attributes can depend on a hidden field, but only for some intakes (a seventh, `transport_assistance`, depends on one in the schema but is always fixed by the intake). Ground truth is a deterministic function of intake and hidden fields; the hidden fields are independent draws from a fixed prior. The agent is a real LLM; only the patient is simulated. A second domain, a travel-booking assistant, has the same structure with different content: five intake answers (trip type, budget, travellers, duration, booking window), four undisclosed traveller facts (loyalty tier, mobility needs, dietary needs, seat preference), and twelve booking attributes, six of which can depend on an undisclosed fact. It was specified and registered before any of its data existed.

Reactions. In the main corpora the reaction to a commitment is its correctness bit flipped with probability $\varepsilon = 0.25$, absent for two attributes the patient never reacts to. In the *LLM-patient* corpus a second model (Llama-3.1-8B) role-plays the patient: it is given its own history but not the clinic's decision, one of three personas (attentive, trusting, anxious), and the agent's commitment as a hedged plain-language sentence, and replies in one or two sentences. A

third model that sees only the sentence and the reply labels it as agreement, contradiction or neither. The agent reads the patient's verbatim replies. An *informed* variant also tells the patient the clinic's decision. A registered gate guards this corpus: a 100-session pilot runs first, and the full 600 sessions run only if contradictions are at least 1.5 times as frequent after wrong commitments as after correct ones.

Development and confirmation. All method development used two development corpora (Llama-3.3-70B, 600 sessions; Llama-3.1-8B, 200). An independent audit of these results found, among other issues, that the agent had been shown a corrupted feedback display (no reaction appeared as acceptance, and below $p = 0.5$ acceptance and contradiction appeared swapped), that resamples covered only five attributes, and that a feedback-free classifier nearly matched the posterior on L. Before generating new data, we committed an analysis plan to the code repository [24]: labels, detectors (V6 primary), every comparison, the statistics, and the wording each outcome licenses. Two amendments, also committed before any confirmatory data existed, added the analyses of Section 7.3, the supervised feedback increment, and the second domain. The confirmatory corpora use a fresh random seed (no session seen in development), show the agent the raw reaction, draw three resamples on every turn from that turn's own context, and cover five agents: Llama-3.3-70B, Llama-3.1-8B, Qwen2.5-72B, GPT-4o-mini and Mistral-Small-3.1-24B, 600 sessions each. Every result below is confirmatory unless marked as development.

Statistics. Splits are 70/30 by session for three split seeds; detectors are fit on the training split and score the test split once; AUROC is averaged over seeds. Differences use a paired bootstrap (2,000 resamples) clustered by session, since test sets of different seeds overlap. Pooled estimates average per-agent differences, with sessions resampled within agent. A difference is called significant when its 95% interval excludes zero; we do not correct for multiplicity and report every registered comparison.

Oracle-information analyses. Because the simulator's rules and hidden-field prior are known, we can compute three exact quantities. The *population risk* $\bar{\eta} = \sum_h P(h) L_U(C,h)$ is the probability that the session's commitments would be wrong for a random user with the same intake. $\eta_\varnothing = P(L_U = 1 \mid X, C)$ adds the reactions the agent was shown (all but the last); by Theorem 1 it bounds every detector computed from the agent's context and its resamples. η_F adds the last reaction and bounds every detector of commitments and reactions. The agent's policy does not involve the hidden facts, so its factors cancel and no agent model is needed. These are not deployable detectors; they locate the information. We also give V6 and ctx-LR(+F) the same oracle knowledge of which commitments are user-dependent in each session (V6-session, LR-session).

Table 1: Per-commitment check of Corollary 1 on user-dependent commitments. e_1, e_0: probability that the three resamples do not all agree, given that the commitment is wrong / right. AUROC for predicting *that* commitment's error.

Agent	wrong	all agree	e_1	e_0	Disagreement	One reaction
Llama-70B	??%	??%	??	??	??	??
Llama-8B	39.7%	49.5%	0.544	0.478	0.555	0.746
Qwen-72B	43.1%	83.1%	0.189	0.154	0.517	0.744
GPT-4o-mini	42.9%	85.6%	0.183	0.115	0.535	0.746
Mistral-24B	40.6%	69.2%	0.344	0.283	0.534	0.746

7 Results

7.1 Sampling cannot see the user

Table 1 shows the mechanism. On commitments whose truth depends on undisclosed facts, agents that are self-consistent repeat the same answer whether it is right or wrong (e_0 and e_1 small and close, so by Corollary 1 the AUROC of disagreement lies within $\frac{1}{2}e_0e_1$ of $\frac{1}{2} + \frac{1}{2}(e_1 - e_0)$), so disagreement carries little information, while one reaction with 25% noise carries a great deal. The development corpora show the same pattern (70B: all three resamples agree on 92.1% of these commitments, disagreement AUROC 0.500, one reaction 0.764). At the session level (Table 2), SelfCheckGPT and semantic entropy score 0.566 [0.535, 0.595] and 0.494 [0.479, 0.510] pooled over agents, both intervals inside the registered band of $[0.40, 0.60]$ (H1): sampling detectors are near chance on user-specific hallucination. They are not near chance everywhere. On the composite label, where many errors are guesses about clinic rules that do vary across resamples, SelfCheckGPT reaches 0.676 [0.645, 0.707], and on AgentBench database questions [17], where errors are computational, it reaches 0.712 with the 70B agent (development). The failure is specific to facts the user never disclosed.

7.2 Detecting user-specific hallucination

Table 2 gives the confirmatory results and Table 3 the registered hypotheses, pooled over agents. The user's implicit reactions add detectable signal beyond population base rates: +.072 [+.054, +.090]* to the posterior (H2, significant for every agent) and +.058 [+.038, +.080]* to a supervised detector (H2b). Detectors that read commitments and reactions beat both sampling detectors by wide margins (H4, H4b). The registered comparison with the strongest context-only detector goes the other way: the posterior does not beat a supervised model that sees only the commitments (+.008 [−.028, +.045], H3), so on this label context alone matches PGE's posterior, and a supervised model given the same commitments and reactions

Table 2: Session AUROC on the user-specific label L_U (mean of three splits; confirmatory corpora, 600 sessions per agent). The composite label L is in the repository. CoVe was run for the 70B agent only.

	Detector	Llama-70B	Llama-8B	Qwen-72B	GPT-4o-mini	Mistral-24B
Model only	SelfCheckGPT	??	.610	.559	.576	.520
	Semantic entropy	??	.533	.485	.486	.474
	CoVe	??	–	–	–	–
Commitments	Mean confidence	??	.512	.522	.465	.454
	ctx-LR	??	.686	.715	.635	.556
	V6−prior	??	.591	.593	.575	.577
Commitments + reactions	Contradicted confident	??	.614	.628	.576	.637
	Graph-feature LR	??	.647	.653	.609	.596
	GCN	??	.574	.663	.597	.569
	Best graphon	??	.531	.506	.521	.409
	ctx-LR+F	??	.725	.748	.668	.682
	PGE posterior (V6)	??	.672	.650	.636	.666
	L_U positive rate	??%	29.7%	39.0%	35.3%	40.3%

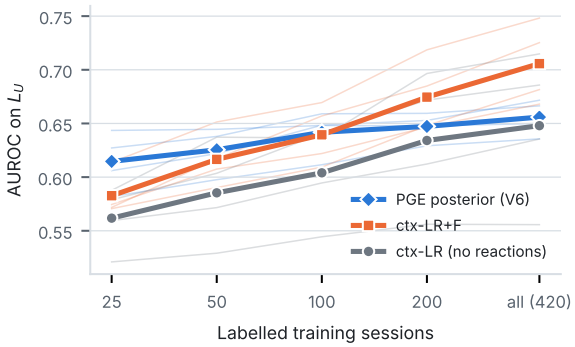


Figure 1: Test AUROC on L_U against the number of labelled training sessions (mean over agents; thin lines: individual agents).

Table 3: Registered hypotheses on L_U, pooled over five agents (stratified session bootstrap; *: 95% CI excludes 0).

	Comparison	Estimate [95% CI]
H2	V6 − V6−prior (feedback increment)	+.072 [+.054, +.090]*
H2b	ctx-LR+F − ctx-LR (feedback increment)	+.058 [+.038, +.080]*
H3	V6 − ctx-LR	+.008 [−.028, +.045]
H4	V6 − SelfCheckGPT	+.090 [+.056, +.123]*
H4	V6 − semantic entropy	+.162 [+.127, +.195]*
H4b	ctx-LR+F − SelfCheckGPT	+.139 [+.101, +.176]*
H4b	ctx-LR+F − semantic entropy	+.211 [+.179, +.242]*
–	V6 − ctx-LR+F	−.050 [−.084, −.019]*

Table 4: Oracle-information analysis on L_U (AUROC). $\bar{\eta}$: population risk of the commitments; $\eta_\varnothing$: Bayes given the conversation the agent saw (Theorem 1's bound for context-and-sample detectors); η_F: Bayes given all reactions. V6-session and LR-session+F are told which commitments are user-dependent in each session.

Agent	$\bar{\eta}$	$\eta_\varnothing$	η_F	V6-session	LR-session+F
Llama-70B	??	??	??	??	??
Llama-8B	.899	.952	.953	.904	.908
Qwen-72B	.854	.930	.932	.879	.851
GPT-4o-mini	.803	.907	.909	.837	.825
Mistral-24B	.781	.890	.891	.847	.814

Pooled: $\eta_F - \bar{\eta} = +.087\,[+.069, +.107]^*$ (H5); V6-session − LR-session+F $= +.017\,[+.000, +.036]^*$ (H6)

is ahead of it when trained on the full split (−.050 [−.084, −.019]*). The ordering depends on how many labelled sessions are available (Figure 1): with 25, the posterior leads on every agent (mean 0.615 against 0.583); the supervised model catches up at about 100 and is ahead with all 0.656 vs. 0.706. Labels for user-specific hallucination require knowing the user's private facts, so small label budgets are the realistic case. On the composite label the posterior beats the context-only model (+.040 [+.015, +.063]*) and ties the supervised model with reactions (+.009 [−.012, +.027]).

7.3 Where the information is

Table 4 locates the signal. The population risk $\bar{\eta}$ alone is far above every sampling detector: much of L_U is visible in *which* values an agent commits to, given the intake. The user's reactions add $\eta_F - \bar{\eta}$ beyond it (H5). Almost all of that information is already in the conversation the agent sees ($\eta_\varnothing$ is close to η_F), so the failure of sampling detectors is not that the information is absent from their input; it is that agreement between resamples does not read it. Given the same information (which commitments concern undisclosed facts), the PGE posterior beats the supervised detector (H6), though by a small margin whose interval nearly reaches zero (+.017 [+.000, +.036]*); it also exceeds the population risk (+.033 [+.010, +.056]*). Without that knowledge, V6 targets a schema-level set of attributes that is only a proxy for the label's support, which costs it 0.181–0.233 AUROC against V6-session: knowing *which* commitments concern undisclosed facts matters as much as how the evidence is combined.

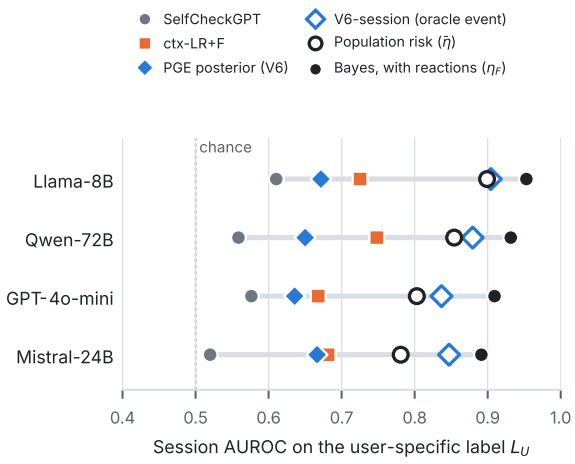


Figure 2: Where the signal is (L_U). Sampling sits near chance; practical detectors on PGE observables in the middle; the population risk $\bar{\eta}$ and the Bayes posterior with reactions η_F (oracle information) bound what is achievable without and with the user.

Table 5: Second domain (travel booking), AUROC on L_U, 600 sessions per agent.

Agent	SelfCheck	ctx-LR	ctx-LR+F	V6	$\bar{\eta}$	η_F
GPT-4o-mini	??	??	??	??	??	??
Llama-8B	??	??	??	??	??	??

7.4 A second domain

[PENDING: second-domain paragraph: does the pattern transfer (sampling near chance, population risk large, reactions add, V6 vs supervised)?]

7.5 Natural-language reactions

The first simulated patient failed its gate. It contradicted 83% of wrong commitments and 79% of correct ones (lift 1.06; 100 sessions, 817 replies), against 75% and 25% for the noisy bit. Reading its replies shows why: the patient voiced doubt about facts it had not been given, and the labeller counted doubt as contradiction. As registered, it was not run at scale. A second version, registered before its data existed (Amendment 5), uses GPT-4o-mini for both roles, lets the patient object only to commitments that conflict with its listed facts, and labels a reply as a contradiction only if it states a conflict. [PENDING: v2 audit (P(-1|wrong), P(-1|right), lift) and, if it passes, detection V6 / V6-prior / ctx-LR / ctx-LR+F on both labels] A simulated user that does not track its own facts produces reactions that no detector can use; the gate is what keeps such a corpus out of the results.

7.6 Choosing what to ask

[PENDING: measured clarification: posterior@1 vs no-question, random, lowest-confidence; posterior@2 vs detailed questions and the full-rerun control; questions per session; primary outcome wrong confident user-dependent commitments]

7.7 Development-phase robustness

On the development corpora we stress-tested the posterior by replacing the reactions seen by the detectors (not the agent) with channels it does not assume: symmetric flips at 0.10–0.40, asymmetric flips, attribute-dependent noise, extra missingness, and a "lazy" user who contradicts 60% of wrong commitments and never confirms correct ones (labels L, ten channels). The symmetric posterior V3 collapses under the lazy user (0.425 for 70B, 0.576 for 8B): it reads the absence of confirmations as evidence of correctness. The general-channel V6 recovers the channel from data and repairs this (0.857 and 0.855). Across all ten channels V6 has the best point estimate on both agents; with session-clustered intervals its lead over the best baseline is significant in six channels for 70B and two for 8B, so the robustness claim is about consistency, not significance. Structure-based detection does not help: graphon likelihoods score between 0.43 and 0.58 on every development corpus and label, even with the best variant chosen on test, and exact inference in a simple cascade model over the dependency graph (errors depend on attribute, bucket, and whether any parent is wrong) is no better than independence (0.732 vs. 0.734 without value conditioning).

8 Limitations

Simulation. The agents are real, but the patient is simulated: by a noisy correctness bit in the main corpora and by an LLM in one corpus. The oracle analyses rely on the simulator's known rules and prior, which a deployment does not have. *Two synthetic domains.* Both private-attribute domains are simulators we designed, with deterministic rules and independent hidden facts; real user populations have correlated, partially self-reported facts. *Labels.* Labels are programmatic and noise-free by construction; there is no human evaluation. *Supervision.* The posterior and the supervised baselines are fit on labelled training sessions; the posterior can run without labels when the channel is known, which we only test in development. *White-box detectors.* We did not run hidden-state probes; Theorem 1 predicts they share the within-context limit, but they could exploit population risk as ctx-LR does. *Theory.* Proposition 1 is exact only under (A1)–(A2) with the true priors and channel, and the detector is a plug-in estimate; Theorem 2 covers reactions regenerated for the detector with the agent's trajectories fixed, not the protocol's own noise, because the agent conditions on the reactions it is shown.

9 Conclusion

When an agent invents a fact about its user, its own samples cannot tell which of the users sharing its context it is wrong about. Some of the risk is visible in what the agent commits to; the rest is visible only through the user's reactions. Making the agent state its assumptions as probabilistic commitments and recording the user's reactions makes both available, and a Bayes analysis in a simulator with known truth shows how much each is worth. [PENDING: one-sentence summary of the confirmatory outcome for the posterior and for clarification]

Reproducibility. Code, all trajectories, the preregistered plan with its amendments and commit history, every analysis script, and a ledger of all runs including discarded variants are at [PENDING: anonymous repository link].

Acknowledgements. [PENDING: AI-assistance disclosure per ICAART policy.]

References

[1] Dana Angluin and Philip Laird. Learning from noisy examples. *Machine Learning*, 2(4):343–370, 1988.

[2] Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

[3] David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2): 265–272, 1953.

[4] Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. INSIDE: LLMs' internal states retain the power of hallucination detection. In *International Conference on Learning Representations*, 2024.

[5] A. P. Dawid and A. M. Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C*, 28(1):20–28, 1979.

[6] Mengyi Deng, Zhiwei Li, Xin Li, Tingyu Zhu, Ying Zhao, Zhijiang Guo, and Wei Wang. Uncertainty-aware clarification in LLM agents with information gain. In *International Conference on Machine Learning*, 2026.

[7] Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. Chain-of-verification reduces hallucination in large language models. In *Findings of ACL*, 2024.

[8] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630: 625–630, 2024.

[9] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[10] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.

[11] Thorsten Joachims, Laura Granka, Bing Pan, Helene Hembrooke, and Geri Gay. Accurately interpreting clickthrough data as implicit feedback. In *Proceedings of the 28th Annual International ACM SIGIR Conference*, 2005.

[12] Adam Tauman Kalai and Santosh S. Vempala. Calibrated language models must hallucinate. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*, 2024.

[13] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.

[14] Johannes Kirmayr, Lukas Stappen, and Elisabeth André. CAR-bench: Evaluating the consistency and limit-awareness of LLM agents under real-world uncertainty. *arXiv preprint arXiv:2601.22027*, 2026.

[15] Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint arXiv:2406.15927*, 2024.

[16] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations*, 2023.

[17] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, et al. AgentBench: Evaluating LLMs as agents. In *International Conference on Learning Representations*, 2024.

[18] Xuannan Liu, Xiao Yang, Zekun Li, Peipei Li, and Ran He. AgentHallu: Benchmarking automated hallucination attribution of LLM-based agents. *arXiv preprint arXiv:2601.06818*, 2026.

[19] László Lovász. *Large Networks and Graph Limits*. American Mathematical Society, 2012.

[20] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of EMNLP*, 2023.

[21] Niels Mündler, Jingxuan He, Slobodan Jenko, and Martin Vechev. Self-contradictory hallucinations of large language models: Evaluation, detection and mitigation. In *International Conference on Learning Representations*, 2024.

[22] Nagarajan Natarajan, Inderjit S. Dhillon, Pradeep Ravikumar, and Ambuj Tewari. Learning with noisy labels. In *Advances in Neural Information Processing Systems*, 2013.

[23] Jerzy Neyman and Egon S. Pearson. On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society A*, 231:289–337, 1933.

[24] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.

[25] Sudha Rao and Hal Daumé III. Learning to ask good questions: Ranking clarification questions using neural expected value of perfect information. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018.

[26] Allen Z. Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, Peng Xu, Leila Takayama, Fei Xia, Jake Varley, Zhenjia Xu, Dorsa Sadigh, Andy Zeng, and Anirudha Majumdar. Robots that ask for help: Uncertainty alignment for large language model planners. In *Conference on Robot Learning (CoRL)*, 2023.

[27] Jost Schatzmann, Karl Weilhammer, Matt Stuttle, and Steve Young. A survey of statistical user simulation techniques for reinforcement-learning of dialogue management strategies. *The Knowledge Engineering Review*, 21(2):97–126, 2006.

[28] Clayton Scott, Gilles Blanchard, and Gregory Handy. Classification with asymmetric label noise: Consistency and maximal denoising. In *Conference on Learning Theory (COLT)*, 2013.

[29] Mrinank Sharma, Meg Tong, Tomasz Korbak, et al. Towards understanding sycophancy in language models. In *International Conference on Learning Representations*, 2024.

[30] Manan Suri, Puneet Mathur, Nedim Lipka, Franck Dernoncourt, Ryan A. Rossi, and Dinesh Manocha. Structured uncertainty guided clarification for LLM agents. *arXiv preprint arXiv:2511.08798*, 2025.

[31] Dennis Ulmer, Martin Gubri, Hwaran Lee, Sangdoo Yun, and Seong Joon Oh. Calibrating large language models using their generations only. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[32] Jerry Wei, Chengrun Yang, Xinying Song, Yifeng Lu, Nathan Hu, Jie Huang, Dustin Tran, Daiyi Peng, Ruibo Liu, Da Huang, Cosmo Du, and Quoc V. Le. Long-form factuality in large language models. In *Advances in Neural Information Processing Systems*, 2024.

[33] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*, 2024.

[34] Michael J. Q. Zhang and Eunsol Choi. Clarify when necessary: Resolving ambiguity through interaction with LMs. In *Findings of NAACL*, 2025.

[35] Muru Zhang, Ofir Press, William Merrill, Alisa Liu, and Noah A. Smith. How language model hallucinations can snowball. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[36] Weichen Zhang, Yiyou Sun, Pohao Huang, Jiayue Pu, Heyue Lin, and Dawn Song. MIRAGE-Bench: LLM agent is hallucinating and where to find them. *arXiv preprint arXiv:2507.21017*, 2025.